

Mathematical and Logical Foundations of Computer Science

Revision Lecture

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What did we cover in this module?

- ▶ Propositional logic
- ▶ Predicate logic

Today

Revision

- ▶ **Propositional Logic**

- ▶ Syntax
- ▶ Natural Deduction proofs
- ▶ Semantics
- ▶ SAT

- ▶ **Predicate Logic**

- ▶ Syntax
- ▶ Natural Deduction proofs
- ▶ Semantics

Propositional Logic – Syntax & Informal Semantics

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We also introduced four connectives:

- ▶ $P \wedge Q$: we have a proof of both P and Q
- ▶ $P \vee Q$: we have a proof of at least one of P and Q
- ▶ $P \rightarrow Q$: if we have a proof of P then we have a proof of Q
- ▶ $\neg P$: stands for $P \rightarrow \perp$

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- ▶ $P \wedge Q \vee R$ means $(P \wedge Q) \vee R$
- ▶ $P \wedge Q \rightarrow Q \wedge P$ means $(P \wedge Q) \rightarrow (Q \wedge P)$

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What are the parse trees of the above formulas?

Propositional Logic – Constructive Natural Deduction

Constructive Natural Deduction rules:

$$\begin{array}{c}
 \frac{\perp}{A} [\perp E] \quad \frac{}{\top} [\top I] \quad \frac{\overline{A}^1 \vdots B}{A \rightarrow B}^1 [\rightarrow I] \quad \frac{A \rightarrow B \quad A}{B} [\rightarrow E] \\
 \\
 \frac{\overline{A}^1 \vdots \perp}{\neg A}^1 [\neg I] \quad \frac{\neg A \quad A}{\perp} [\neg E] \\
 \\
 \frac{A}{A \vee B} [\vee I_L] \quad \frac{A}{B \vee A} [\vee I_R] \quad \frac{A \vee B \quad A \rightarrow C \quad B \rightarrow C}{C} [\vee E] \\
 \\
 \frac{A \quad B}{A \wedge B} [\wedge I] \quad \frac{A \wedge B}{B} [\wedge E_R] \quad \frac{A \wedge B}{A} [\wedge E_L]
 \end{array}$$

Propositional Logic – Classical Reasoning

Classical Natural Deduction includes all the Constructive Natural Deduction rules, plus:

$$\frac{}{A \vee \neg A} \quad [LEM] \qquad \frac{\neg \neg A}{A} \quad [DNE]$$

Propositional Logic – Natural Deduction

Typical proof patterns when doing proofs bottom-up:

Propositional logic:

- ▶ $\vee$: first $[\vee E]$ then $[\vee I]$
- ▶ $\wedge$: first $[\wedge I]$ then $[\wedge E]$
- ▶ $\rightarrow$: first $[\rightarrow I]$ then $[\rightarrow E]$
- ▶ $\neg$: first $[\neg I]$ then $[\neg E]$

Predicate logic:

- ▶ $\exists$: first $[\exists E]$ then $[\exists I]$
- ▶ $\forall$: first $[\forall I]$ then $[\forall E]$

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$$\frac{\frac{\frac{}{A \rightarrow C} 1 \quad \frac{\frac{}{A \wedge B} 3}{A} [\wedge E_L]}{C} [\rightarrow E] \quad \frac{}{\neg C} 2}{\perp} [\neg E] \quad \frac{}{\neg(A \wedge B)} 3 [\neg I] \quad \frac{}{\neg C \rightarrow \neg(A \wedge B)} 2 [\rightarrow I] \quad \frac{}{(A \rightarrow C) \rightarrow \neg C \rightarrow \neg(A \wedge B)} 1 [\rightarrow I]$$

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- ▶ A is **valid** if $\phi(A) = \mathbf{T}$ for all possible valuations ϕ

Truth Tables

We can use **truth tables** to check whether propositions are valid:

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

A	$\neg A$
T	F
F	T

A proposition is (semantically) valid if the last column in its truth table only contains **T**

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These techniques can be used to prove the validity of propositions:

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- ▶ A is satisfiable
- ▶ A is not falsifiable

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Is $(A \rightarrow C) \rightarrow \neg C \rightarrow \neg(A \wedge B)$ valid/satisfiable?

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- ▶ Is it satisfiable?
- ▶ **Satisfiable** by setting $A = \mathbf{T}$, $B = \mathbf{F}$ and $C = \mathbf{T}$
- ▶ Known as **CNF Satisfiability** or simply **SAT**

Predicate Logic – Syntax

The syntax of predicate logic is defined by the following grammar:

$$t ::= x \mid f(t, \dots, t)$$

$$P ::= p(t, \dots, t) \mid \neg P \mid P \wedge P \mid P \vee P \mid P \rightarrow P \mid \forall x.P \mid \exists x.P$$

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where:

- ▶ x ranges over variables
- ▶ f ranges over function symbols
- ▶ $f(t_1, \dots, t_n)$ is a well-formed term only if f has arity n
- ▶ p ranges over predicate symbols
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The pair of a collection of function symbols, and a collection of predicate symbols, along with their arities, is called a **signature**.

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The scope of a quantifier extends as far right as possible. E.g., $P \wedge \forall x.p(x) \vee q(x)$ is read as $P \wedge \forall x.(p(x) \vee q(x))$

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$(p(t_1, \dots, t_n))[x \backslash t]$	$=$	$p(t_1[x \backslash t], \dots, t_n[x \backslash t])$
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The additional **conditions** ensure that **free variables do not get captured**.

These conditions can always be met by silently renaming bound variables before substituting.

Predicate Logic – Constructive Natural Deduction

Natural Deduction rules for the propositional connectives:

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 \frac{\perp}{A} [\perp E] \qquad \frac{}{\top} [\top I] \qquad \frac{\overline{A}^1 \vdots B}{A \rightarrow B}^1 [\rightarrow I] \qquad \frac{A \rightarrow B \quad A}{B} [\rightarrow E] \\
 \\
 \frac{\overline{A}^1 \vdots \perp}{\neg A}^1 [\neg I] \qquad \frac{\neg A \quad A}{\perp} [\neg E] \\
 \\
 \frac{A}{A \vee B} [\vee I_L] \qquad \frac{A}{B \vee A} [\vee I_R] \qquad \frac{A \vee B \quad A \rightarrow C \quad B \rightarrow C}{C} [\vee E] \\
 \\
 \frac{A \quad B}{A \wedge B} [\wedge I] \qquad \frac{A \wedge B}{B} [\wedge E_R] \qquad \frac{A \wedge B}{A} [\wedge E_L]
 \end{array}$$

Predicate Logic – Constructive Natural Deduction

Natural Deduction rules for quantifiers:

$$\begin{array}{c}
 \frac{P[x \backslash y]}{\forall x.P} \quad [\forall I] \qquad \frac{\forall x.P}{P[x \backslash t]} \quad [\forall E] \qquad \frac{P[x \backslash t]}{\exists x.P} \quad [\exists I] \qquad \frac{\exists x.P \quad \begin{array}{c} \overline{P[x \backslash y]}^1 \\ \vdots \\ Q \end{array}}{Q}^1 \quad [\exists E]
 \end{array}$$

Condition:

- ▶ for $[\forall I]$: y must not be free in any not-yet-discharged hypothesis or in $\forall x.P$
- ▶ for $[\forall E]$: $\mathbf{fv}(t)$ must not clash with $\mathbf{bv}(P)$
- ▶ for $[\exists I]$: $\mathbf{fv}(t)$ must not clash with $\mathbf{bv}(P)$
- ▶ for $[\exists E]$: y must not be free in Q or in not-yet-discharged hypotheses or in $\exists x.P$

Predicate Logic – Classical Reasoning

Classical Natural Deduction for Predicate Logic

Add the rules:

$$\frac{}{A \vee \neg A} \quad [LEM] \qquad \frac{\neg \neg A}{A} \quad [DNE]$$

Predicate Logic – Example of a Proof

Prove $(S_1) \rightarrow (S_2) \rightarrow \forall x.\text{even}(x) \rightarrow \neg\text{odd}(\text{succ}(\text{succ}(x)))$ where
 $S_1 = \forall x.\text{even}(x) \rightarrow \text{even}(\text{succ}(\text{succ}(x)))$ and
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$$\begin{array}{c}
 \frac{}{S_1} \quad 1 \qquad \frac{}{S_2} \quad 2 \\
 \frac{E(x) \rightarrow E(s(s(x)))}{E(s(s(x)))} \quad [\forall E] \quad \frac{E(x)}{E(x)} \quad 3 \qquad \frac{O(s(s(x))) \rightarrow \neg E(s(s(x)))}{O(s(s(x))) \rightarrow \neg E(s(s(x)))} \quad [\forall E] \quad \frac{O(s(s(x)))}{O(s(s(x)))} \quad 4 \\
 \frac{}{E(s(s(x)))} \quad [\rightarrow E] \qquad \frac{}{\neg E(s(s(x)))} \quad [\rightarrow E] \\
 \frac{}{\perp} \quad 4 \quad [\neg E] \\
 \frac{}{\neg O(s(s(x)))} \quad 4 \quad [\neg I] \\
 \frac{}{E(x) \rightarrow \neg O(s(s(x)))} \quad 3 \quad [\rightarrow I] \\
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 \frac{}{(S_2) \rightarrow \forall x.E(x) \rightarrow \neg O(s(s(x)))} \quad 2 \quad [\rightarrow I] \\
 \frac{}{(S_1) \rightarrow (S_2) \rightarrow \forall x.E(x) \rightarrow \neg O(s(s(x)))} \quad 1 \quad [\rightarrow I]
 \end{array}$$

Where O stands for **odd**; E stands for **even**; and s stands for **succ**

Predicate Logic – Semantics

Models: a model provides the interpretation of all symbols

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Given a **signature** $\langle \langle f_1^{k_1}, \dots, f_n^{k_n} \rangle, \langle p_1^{j_1}, \dots, p_m^{j_m} \rangle \rangle$

- ▶ of function symbols f_i of arity k_i , for $1 \leq i \leq n$
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- ▶ of a non-empty domain D
- ▶ interpretations $\mathcal{F}_{f_i}$ for function symbols f_i
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Models of predicate logic replace **truth assignments** for propositional logic

Variable valuations:

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- ▶ that map variables to D
- ▶ i.e., a mapping of the form $x_1 \mapsto d_1, \dots, x_n \mapsto d_n$

Predicate Logic – Semantics

Given a **model** M with domain D and a **variable valuation** v :

- ▶ $\llbracket t \rrbracket_v^M$ gives meaning to the term t w.r.t. M and v
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Meaning of terms:

- ▶ $\llbracket x \rrbracket_v^M = v(x)$
- ▶ $\llbracket f(t_1, \dots, t_n) \rrbracket_v^M = \mathcal{F}_f(\langle \llbracket t_1 \rrbracket_v^M, \dots, \llbracket t_n \rrbracket_v^M \rangle)$

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Meaning of formulas:

- ▶ $\models_{M,v} \top$ and $\not\models_{M,v} \perp$
- ▶ $\models_{M,v} p(t_1, \dots, t_n)$ iff $\langle \llbracket t_1 \rrbracket_v^M, \dots, \llbracket t_n \rrbracket_v^M \rangle \in \mathcal{R}_p$
- ▶ $\models_{M,v} \neg P$ iff $\not\models_{M,v} P$
- ▶ $\models_{M,v} P \wedge Q$ iff $\models_{M,v} P$ and $\models_{M,v} Q$
- ▶ $\models_{M,v} P \vee Q$ iff $\models_{M,v} P$ or $\models_{M,v} Q$
- ▶ $\models_{M,v} P \rightarrow Q$ iff $\models_{M,v} Q$ whenever $\models_{M,v} P$
- ▶ $\models_{M,v} \forall x.P$ iff for every $d \in D$ we have $\models_{M,(v,x \mapsto d)} P$
- ▶ $\models_{M,v} \exists x.P$ iff there exists a $d \in D$ such that $\models_{M,(v,x \mapsto d)} P$

Predicate Logic – Soundness & Completeness

Natural Deduction for Predicate Logic is

- ▶ **sound**, i.e., if $\vdash A$ then $\models A$, and
- ▶ **complete**, i.e., if $\models A$ then $\vdash A$

w.r.t. the **model semantics of Predicate Logic**

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Consider the signature: $\langle\langle\rangle, \langle\text{even}^1, \text{odd}^1\rangle\rangle$

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Examples of models of P ?

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- ▶ $\langle \mathbb{N}, \diamond, \langle \{ \langle n \rangle \mid n \text{ is even} \}, \{ \langle n \rangle \mid n \text{ is even} \} \rangle \rangle$
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- ▶ $\langle \mathbb{N}, \diamond, \langle \{ \langle 1 \rangle \}, \{ \langle 1 \rangle \} \rangle \rangle$
- ▶ $\langle \mathbb{N}, \diamond, \langle \emptyset, \{ \langle 0 \rangle \} \rangle \rangle$

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